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Efficacy and Safety of Lorundrostat in Uncontrolled Hypertension: A Systematic Review and Meta-Analysis

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Abstract

This systematic review and meta-analysis evaluated the efficacy and safety of lorundrostat in adults with uncontrolled hypertension. Following PRISMA guidelines and PROSPERO registration (CRD420251088503), five databases were systematically searched through July 2025 for randomized controlled trials comparing lorundrostat with placebo in this population. The primary outcome was change in systolic blood pressure (SBP), while secondary outcomes included diastolic blood pressure, severe BP events, and adverse effects. Three RCTs comprising 1568 participants across 10 study arms were included. Lorundrostat significantly reduced 24-h ambulatory SBP (mean difference [MD]: −7.45 mmHg; 95% CI: −12.54 to −2.36; $p = 0.0041$; $p^2 = 0\%$) and diastolic BP (MD: −3.49 mmHg; 95% CI: −5.56 to −1.41; $p = 0.0010$; $I^2 = 0\%$). While office SBP showed a non-significant reduction in the primary analysis (MD: −13.55 mmHg; $p = 0.077$; $I^2 = 94\%$), it became statistically significant in a sensitivity analysis (MD: −9.08 mmHg; $p < 0.0001$). Lorundrostat also significantly lowered the risk of severely elevated BP events (odds ratio [OR]: 0.37; 95% CI: 0.17–0.81; $p = 0.028$). Adverse effects included an increased risk of hyperkalemia (OR: 3.22; $p < 0.001$) and hyponatremia (OR: 2.16; $p = 0.037$), with no significant difference in serious adverse events between groups. In conclusion, lorundrostat demonstrates significant reductions in both ambulatory and diastolic BP in patients with uncontrolled hypertension, with a generally tolerable safety profile. Hyperkalemia and hyponatremia remain notable risks. Further long-term trials are warranted to validate its sustained efficacy and safety.

1 | Introduction

Hypertension remains the leading modifiable risk factor for cardiovascular disease (CVD) and all-cause mortality worldwide [1]. According to the Global Burden of Disease 2023 update, elevated blood pressure contributed to an estimated 10.8 million deaths [2]. Uncontrolled hypertension continues to be a serious global burden on cardiovascular health as well. Its prevalence among adults with hypertension is estimated at 56.3% in the United States, based on NHANES 2017–2018 data [3]. Although a

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wide variety of antihypertensive drugs are available, control rates are far from optimal [4].

Uncontrolled hypertension is defined as systolic blood pressure ≥ 130 mm Hg or diastolic blood pressure ≥ 80 mm Hg in patients receiving antihypertensive treatment [5]. One of the key contributors to uncontrolled hypertension is inappropriate aldosterone activity [6, 7]. Both primary and secondary causes of aldosteronism are associated with persistent elevation of blood pressure [8]. Guideline-recommended therapies for uncontrolled hypertension include angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), calcium channel blockers (CCBs), and thiazide diuretics as initial drug choices, with mineralocorticoid receptor antagonists recommended as add-on therapy when blood pressure remains uncontrolled [9]. However, their use is limited by frequent adverse effects [10]. Therefore, there is growing interest in directly suppressing aldosterone biosynthesis using aldosterone synthase inhibitors (ASIs). Selective ASIs may effectively lower blood pressure while minimizing cortisol suppression, preserving adrenal function, and reducing endocrine-related adverse effects [11, 12].

Lorundrostat is a novel, orally active, selective ASI. Prior studies have demonstrated that it effectively lowers blood pressure, particularly in patients with aldosterone-driven hypertension, with a favorable safety and tolerability profile [13, 14]. A recent meta-analysis by Marzano et al. synthesized early trial data on ASIs, including lorundrostat, baxdrostat, and osilodrostat. However, the included agents varied in mechanism and selectivity, and the studies were short in duration, making it difficult to isolate the clinical utility of lorundrostat [12, 15, 16]. To address this gap, our systematic review and meta-analysis aim to provide targeted, up-to-date evidence on the efficacy and safety of lorundrostat in patients with uncontrolled hypertension.

2 | Methods

2.1 | Data Sources and Search Strategy

In this study, the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines were followed [17]. The meta-analysis was registered in the PROSPERO international prospective register of systematic reviews (Registration No: CRD420251088503). No amendments were made after protocol registration.

A comprehensive literature search from inception through August 2025 was conducted without language restrictions using five databases, including PubMed, Embase, Scopus, ClinicalTrials.gov, and the Cochrane Library. A combination of keywords and MeSH terms, including "uncontrolled hypertension," "Lorundrostat," and "clinical trials," were used with Boolean operators ("AND", "OR") to refine the search. In addition, the bibliographic sections of the selected articles from the online search were manually screened for any other relevant study. The search strategy is available in Table S1.

2.2 | Study Selection and Selection Criteria

Studies were included if they met the following criteria: (1) randomized controlled trial (RCT) design, (2) enrolled patients with uncontrolled hypertension, and (3) compared lorundrostat with placebo in terms of efficacy and safety. Studies, irrespective of duration of intervention and geography, were included as long as they provided relevant information. We restricted our search to studies in human subjects and did not include articles with no full text available. Case reports, case series, editorials, and review articles were also excluded.

2.3 | Data Extraction and Quality Assessment

Results were exported to Covidence, and duplicates were removed by algorithmic deduplication as well as manually [18]. Two authors independently reviewed the titles and abstracts of the identified studies and excluded any article that did not answer the research query or did not follow predefined selection criteria. Full texts of potentially eligible articles were assessed for relevance and eligibility based on predefined criteria. A third investigator was consulted to resolve any discrepancies. The primary outcome was the mean change in the office systolic blood pressure, whereas the secondary outcomes included the risk of any adverse events (AEs), serious AEs, hyperkalemia, and hypotension. In addition, data were extracted on study characteristics (publication year, country, study design), participant characteristics (sample size, mean age, sex distribution, baseline blood pressure, body mass index), and trial features such as duration of follow-up and intervention details. The detailed definitions and explanation of included outcomes are described in Table S2.

The risk of bias was assessed using the Cochrane Risk-of-Bias 2 (RoB-2) tool, which evaluates five domains related to trial design, conduct, and reporting [19]. The risk of bias assessment is presented in Figure S1. In addition, the certainty of evidence for each outcome was assessed using the GRADE approach, and the summary of findings is provided in Table S3. For transparent reporting, the completed PRISMA 2020 checklist is provided in Table S4.

2.4 | Statistical Analysis

R version 4.4.2 (2024-10-31) was used to perform all statistical analyses using the Mantel–Haenszel random-effects model with generic inverse variance to account for clinical heterogeneity. The results of the studies were presented as mean and standard deviations for continuous variables and risk or odds ratios (RR/OR) for dichotomous variables with 95% confidence intervals (CI). A p value < 0.05 was considered statistically significant. All results were pooled using a random-effects model. To avoid unit-of-analysis errors, we divided the shared control group across independent intervention arms. This enabled separate comparisons and exploration of intervention-related heterogeneity. Forest plots were created to assess and visualize the results. Heterogeneity across studies was calculated using the I^2 statistic, with values of 0%, 25%, 50%, and 75% indicating absent, low, moderate, and high heterogeneity, respectively [20]. To assess the robustness of our findings, we performed a leave-one-out

sensitivity analysis, in which the meta-analysis was repeated by sequentially omitting one study at a time to evaluate the influence of individual studies on the pooled effect size. Publication bias was planned to be assessed using Egger's test if ten or more studies were available [21]. To assess the robustness of our findings, we performed a leave-one-out (LOO) sensitivity analysis, in which the meta-analysis was repeated by sequentially omitting one study at a time to evaluate the influence of individual studies on the pooled effect size.

3 | Results

3.1 | Literature Search Results and Study Characteristics

Initial search of the electronic databases yielded 393 potential studies. After screening and application of eligibility criteria, three randomized controlled trials (RCTs) were included in the final pooled analysis [14, 22, 23]. The study selection process is illustrated in the PRISMA flow chart (Figure 1).

The three included RCTs were published between 2023 and 2025. Target-HTN and Advance-HTN were conducted in the United States, whereas Launch-HTN was a multinational trial across 13 countries. Follow-up durations ranged from 8 to 12 weeks. Across these three trials, a total of 1568 patients with uncontrolled hypertension were enrolled, 1165 assigned to lorundrostat intervention arms and 403 to control groups. These three trials comprised a total of 10 study arms evaluating varying dosing regimens of lorundrostat, including both fixed-dose and dose-titrated treatment groups. The proportion of male participants ranged from 32.3% to 60.0% across study arms. Baseline body mass index (BMI) averaged between 30.4 and 33.0 kg/m^2. Office systolic and diastolic blood pressures at screening ranged from 139.8 to 149.0 mmHg and 78.5 to 87.8 mmHg, respectively. Where reported, estimated glomerular filtration rate (eGFR) ranged from 76.4 to 92.8 mL/min/1.73 m^2. The prevalence of diabetes mellitus varied from 26.7% to 51.6% across study arms. Nearly all patients were either on two-drug or three-drug antihypertensive regimens at baseline. Baseline characteristics of all studies are summarized in Table 1.

3.2 | Results of Meta-Analysis

The pooled results of our meta-analysis are presented in Figures 2–15. Forest plots illustrating the effect sizes under random-effects models are provided for each outcome.

3.2.1 | Primary Outcomes

3.2.1.1 | Change in Systolic Blood Pressure (SBP).

a) Change in Office SBP

7 study arms derived from 2 RCTs (lorundrostat: $n = 942$; control: $n = 300$) reported the change in office SBP from baseline to the end of treatment period. Lorundrostat reduced office SBP compared to control; however, this did not reach statistical significance (MD −13.55 mmHg, 95% CI: −28.57 to 1.47; $p = 0.077$) (Figure 2). Substantial heterogeneity was observed ($I^2 = 94\%$). A leave-one-out sensitivity analysis identified the LAUNCH 50 mg adjustment-dose study arm as a key source of inconsistency; omitting it reduced heterogeneity to 0% and yielded a statistically significant MD of −9.08 mmHg (95% CI: −13.20 to −4.95; $p < 0.0001$), showing that lorundrostat reduced office SBP and supporting the robustness of the overall effect (Figure 3).

A supporting single-arm continuous analysis of nine lorundrostat dosing cohorts derived from 3 RCTs ($n = 1067$) showed a pooled mean office SBP reduction of 13.29 mmHg (95% CI: −16.18 to −10.41; $p < 0.0001$) (Figure 4). Despite moderate heterogeneity ($I^2 = 67.7\%$), all dosing cohorts showed reductions in SBP, supporting lorundrostat's antihypertensive effect.

b) 24-h Ambulatory SBP

2 study arms derived from 1 RCT (lorundrostat: $n = 188$; control: $n = 95$) showed a significant reduction in 24-h ambulatory (average) SBP with lorundrostat (MD: −7.45 mmHg; 95% CI: −12.54 to −2.36; $p = 0.0041$). No heterogeneity was detected ($I^2 = 0\%$) (Figure 5).

3.2.1.2 | Change in Office Diastolic Blood Pressure (DBP). 6 study arms derived from 2 RCTs (lorundrostat: $n = 942$; control: $n = 300$) demonstrated a statistically significant reduction in office diastolic blood pressure from baseline to end of treatment, favouring lorundrostat, with a pooled mean difference of −3.49 mmHg (95% CI: −5.56 to −1.41; $p = 0.0010$). No statistical heterogeneity was observed across studies ($I^2 = 0\%$) (Figure 6).

3.2.1.3 | Risk of Severely Elevated Blood Pressure Events. 4 study arms derived from 2 RCTs (lorundrostat: $n = 997$, 16 events; control: $n = 365$, 15 events) showed that lorundrostat was associated with a statistically significant reduction in severely elevated blood pressure events (Odds Ratio (OR): 0.37; 95% CI: 0.17 to 0.81; $p = 0.028$). No heterogeneity was observed ($I^2 = 0\%$) (Figure 7).

3.2.2 | Adverse (Safety) Outcomes

Across the included trials, lorundrostat was associated with a few adverse effects, with some showing statistically significant increases in risk while others did not differ meaningfully from control.

3.2.2.1 | Hyperkalemia. Hyperkalemia was the most prominent adverse effect, with a statistically significant higher risk observed in the lorundrostat group (52/1161) compared with control (1/401) across 10 study arms from 3 RCTs; the pooled OR was 3.22 (95% CI: 2.01 to 5.15; $p < 0.001$) with no observed heterogeneity ($I^2 = 0\%$) (Figure 8).

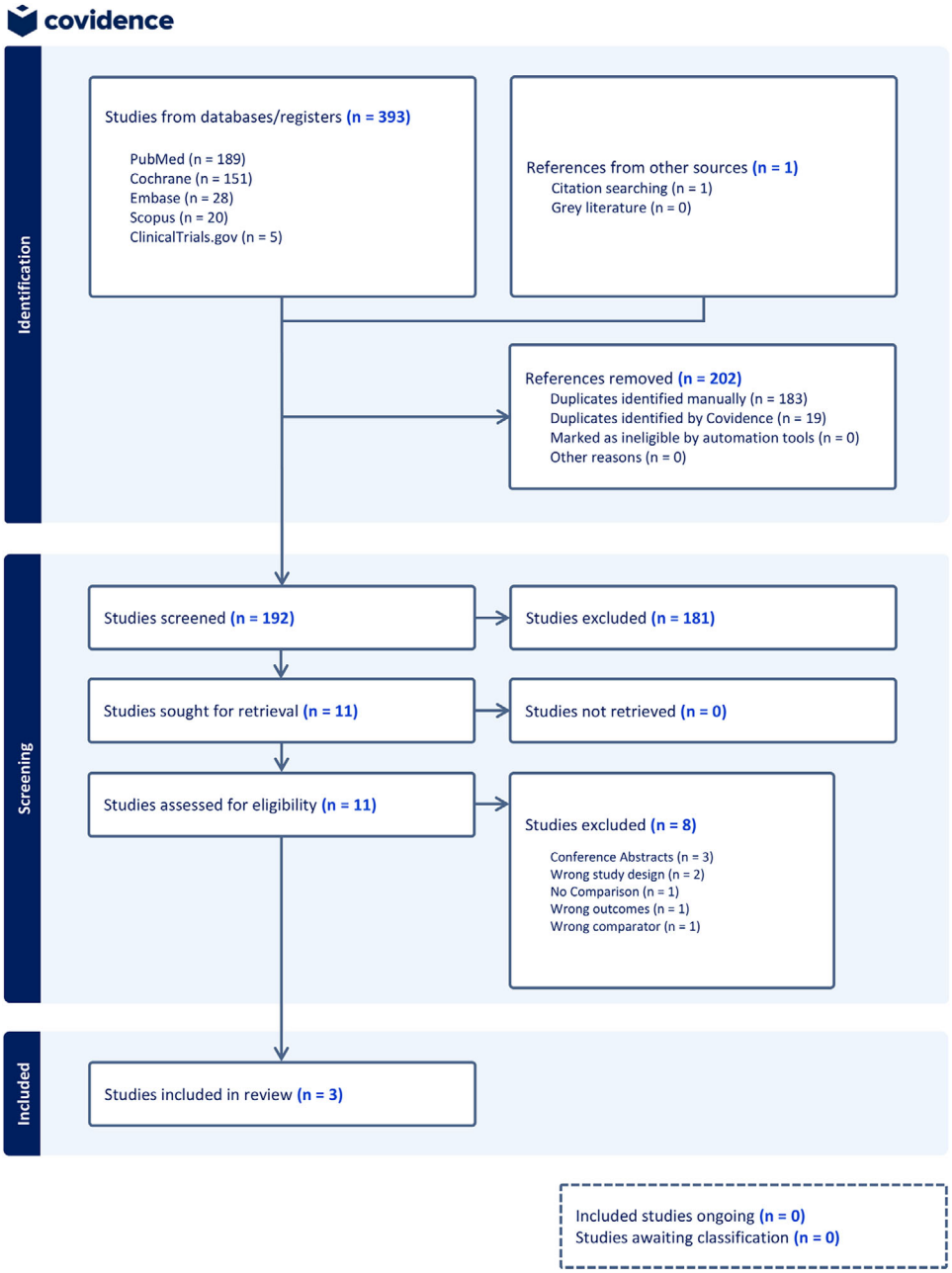

FIGURE 1 | Preferred reporting items for systematic reviews and meta-analyses (PRISMA) flowchart illustrating the study selection process.

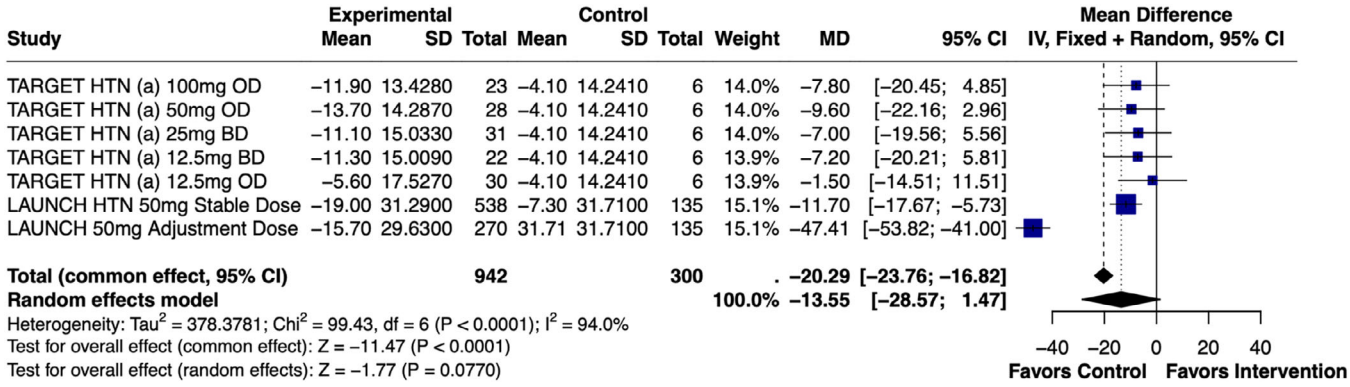

FIGURE 2 | Forest plot of change in office systolic blood pressure with lorundrostat versus control. IV, inverse variance; SD, standard deviation.

TABLE 1 | Baseline characteristics of the included studies.

Trial name (year)	Intervention	Control	Dose subgroup	Sample size (*n*)		Males- *n* (%)		BMI (Kg/m^2)		Baseline Office BP at screening SBP/DBP (mm of Hg)		24-hr average ambulatory BP at randomization (mm of Hg)		eGFR at randomization (mL/min/1.73 m^2)		Diabetes mellitus-*n* (%)	
				I	*C*	*I*	*C*	*I*	*C*	*I*	*C*	*I*	*C*	*I*	*C*	*I*	*C*
Target-HTN (2023) (cohort 1) [14]	Lorundrostat	Placebo	100 mg (OD)	30	30	12 (40.0)	13 (43.3)	30.4 ± 5.5	31.9 ± 5.0	142.2 ± 13.4/78.5 ± 10.0	142.9 ± 10.7/83.8 ± 9.5	—	—	77.4 ± 14.0	81.6 ± 17.3	8 (26.7)	14 (46.7)
	Lorundrostat		50 mg (OD)	28		13 (46.4)		32.0 ± 5.0		140.0 ± 12.1/84.7 ± 7.0		—	—	77.2 ± 14.1		8 (28.6)	
	Lorundrostat		25 mg (BD)	30		11 (36.7)		30.6 ± 5.5		142.8 ± 13.1/80.1 ± 9.3		—	—	80.9 ± 12.4		11 (36.7)	
	Lorundrostat		12.5 mg (BD)	22		8 (36.4)		32.0 ± 5.2		142.6 ± 13.3/81.6 ± 9.4		—	—	81.7 ± 16.3		9 (40.9)	
	Lorundrostat		12.5 mg (OD)	23		11 (47.8)		30.6 ± 4.9		142.9 ± 13.7/80.3 ± 12.0		—	—	77.9 ± 18.7		11 (47.8)	
Target-HTN (2023) (cohort 2) [14]	Lorundrostat	Placebo	100 mg (OD)	31	6	10 (32.3)	2 (33.3)	30.5 ± 4.4	32.0 ± 3.9	139.8 ± 9.1/78.6 ± 10.0	135.3 ± 5.5/81.5 ± 7.9	—	—	79.9 ± 13.1	83.9 ± 18.6	16 (51.6)	2 (33.3)
Advanced-HTN (2025) [22]	Lorundrostat	Placebo	50 mg (Stable Dose)	94	95	56 (60)	62 (65)	31.2 ± 4.6	32.2 ± 4.8	141.8 ± 14.4/84.3 ± 9.4	(141.7 ± 14.0)/85.5 ± 10.4	(140.5±11.3/ 85.5±8.9)	(141.1±9.3/ 86.8±8.8)	76.6 ± 17.8	73.6 ± 18.0	39 (41)	34 (36)
	Lorundrostat		50 mg (Adjustment Dose)	96		54 (56)		32.4 ± 5.4		143.5 ± 12.8/85.6 ± 10.2)		(141.4±11.5/ 86.5±9.4)		76.4 ± 19.4		46 (48)	
Launch-HTN (2025) [23]	Lorundrostat	Placebo	50 mg (Stable Dose)	541	272	294 (54.3)	139 (51.1)	33.0 ± 6.9	32.6 ± 7.4	149.0 ± 12.2/87.8 ± 9.0	148.8 ± 12.2/87.1 ± 9.1	—	—	90.1 ± 17.8	91.2 ± 16.4	173 (32.2)	89 (33.0)
	Lorundrostat		50 mg (Adjustment Dose)	270		142 (52.6)		32.8 ± 6.7		146.6 ± 11.1/86.3 ± 9.2		—	—	92.8 ± 17.6		76 (28.2)	

Note: All the continuous variables are presented as mean ± standard deviation (SD); "-" indicates data not reported.

Abbreviations: BD, twice daily; BMI, body mass index; BP, blood pressure; C, control; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; I, intervention; OD, once daily; SBP, systolic blood pressure.

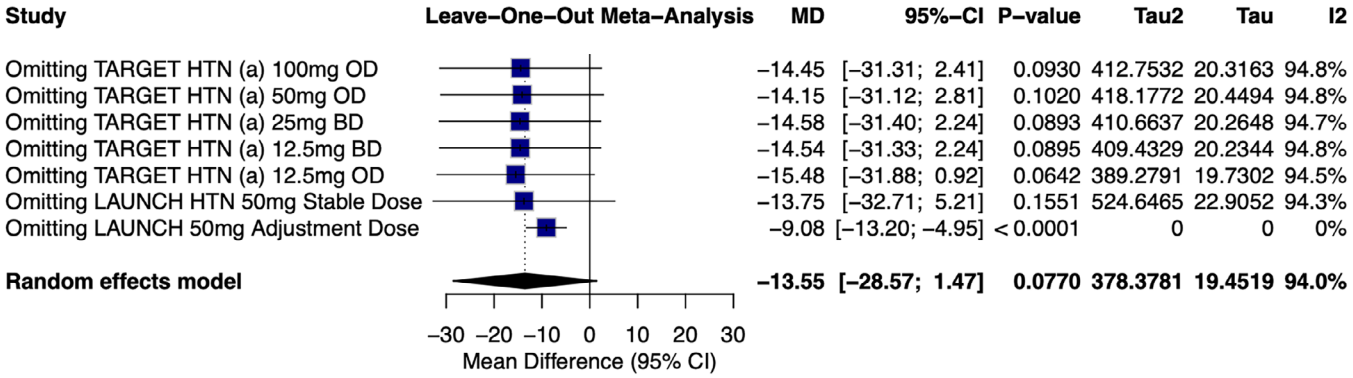

FIGURE 3 | Leave-one-out sensitivity analysis of the change in office systolic blood pressure with lorundrostat versus control. MD, mean difference.

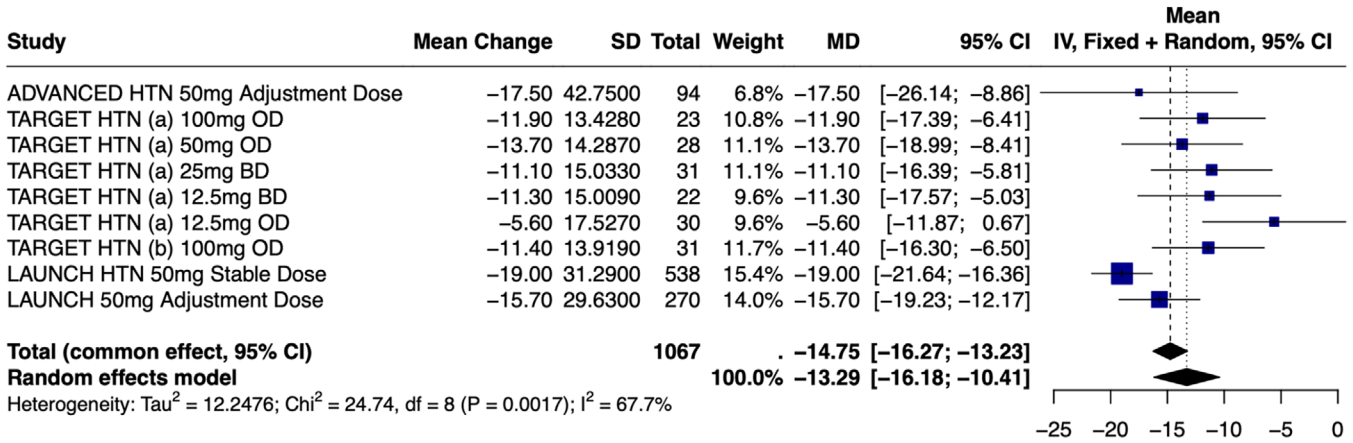

FIGURE 4 | Forest plot of change in office systolic blood pressure in lorundrostat (single arm analysis). IV, inverse variance; MD, mean difference; SD, standard deviation.

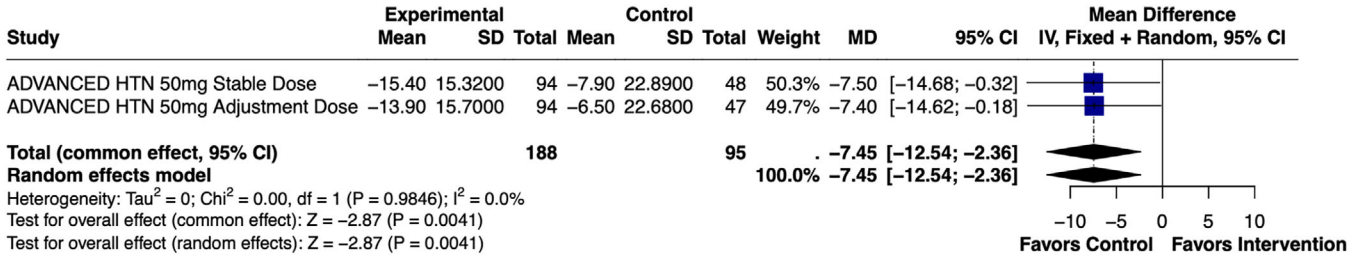

FIGURE 5 | Forest plot of change in 24 h ambulatory systolic blood pressure with lorundrostat versus control. IV, inverse variance; MD, mean difference; SD, standard deviation.

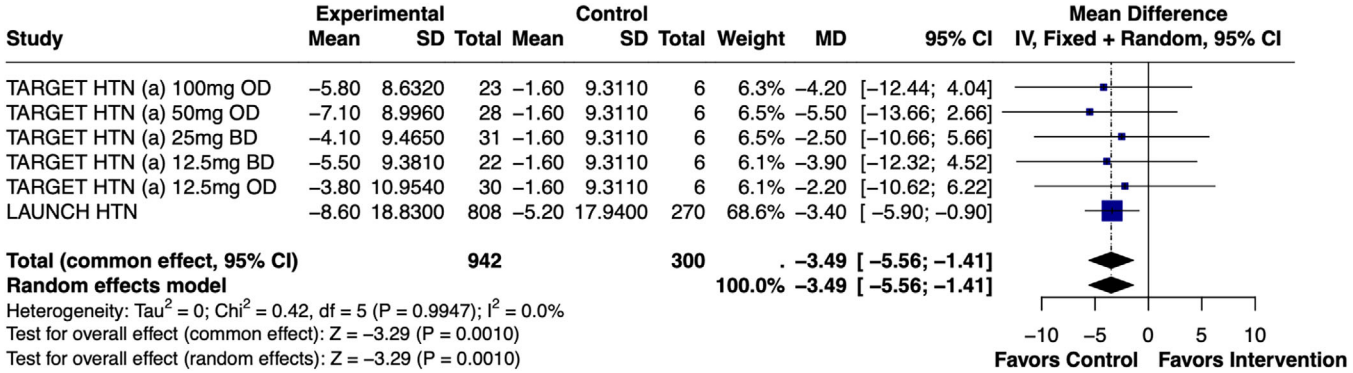

FIGURE 6 | Forest plot of change in office diastolic blood pressure with lorundrostat versus control. IV, inverse variance; MD, mean difference; SD, standard deviation.

Study	Intervention Events	Intervention Total	Control Events	Control Total	Weight (random)	OR	95% CI
ADVANCED HTN 50mg Stable Dose	1	94	2	48	9.6%	0.25	[0.02; 2.80]
ADVANCED HTN 50mg Adjustment Dose	3	95	2	47	16.9%	0.73	[0.12; 4.55]
LAUNCH HTN 50mg Stable Dose	10	538	6	135	53.0%	0.41	[0.15; 1.14]
LAUNCH 50mg Adjustment Dose	2	270	5	135	20.6%	0.19	[0.04; 1.01]
Total (common effect, 95% CI)	**16**	**997**	**15**	**365**	**.**	**0.37**	**[0.17; 0.78]**
Total (random effect, 95% CI)					**100.0%**	**0.37**	**[0.17; 0.81]**

Heterogeneity: $Tau^2 = 0$; $Chi^2 = 1.27$, df = 3 (P = 0.7373); $I^2 = 0.0\%$
Test for overall effect (random effects): $t_3 = -4.02$ (P = 0.028)

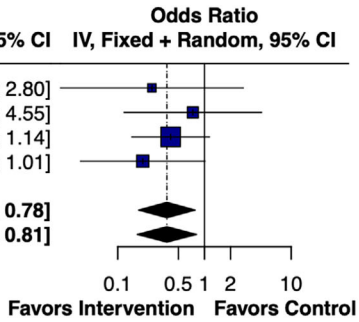

FIGURE 7 | Forest plot of risk of severely elevated blood pressure with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

Study	Intervention Events	Intervention Total	Control Events	Control Total	Weight (random)	OR	95% CI
ADVANCED HTN 50mg Stable Dose	5	94	0	48	9.6%	5.96	[0.32; 110.09]
ADVANCED HTN 50mg Adjustment Dose	8	95	0	47	9.8%	9.23	[0.52; 163.41]
TARGET HTN (a) 100mg OD	6	30	0	6	9.0%	3.45	[0.17; 69.53]
TARGET HTN (a) 50mg OD	2	28	0	6	8.2%	1.23	[0.05; 28.77]
TARGET HTN (a) 25mg BD	3	30	0	6	8.5%	1.65	[0.08; 36.15]
TARGET HTN (a) 12.5mg BD	3	22	0	6	8.5%	2.33	[0.11; 51.45]
TARGET HTN (a) 12.5mg OD	4	23	0	6	8.7%	3.00	[0.14; 63.58]
TARGET HTN (b) 100mg OD	3	31	0	6	8.6%	1.60	[0.07; 34.85]
LAUNCH HTN 50mg Stable Dose	11	538	1	135	19.2%	2.80	[0.36; 21.85]
LAUNCH 50mg Adjustment Dose	7	270	0	135	9.9%	7.71	[0.44; 136.07]
Total (common effect, 95% CI)	**52**	**1161**	**1**	**401**	**.**	**3.22**	**[1.31; 7.92]**
Total (random effect, 95% CI)					**100.0%**	**3.22**	**[2.01; 5.15]**

Heterogeneity: $Tau^2 = 0$; $Chi^2 = 1.84$, df = 9 (P = 0.9937); $I^2 = 0.0\%$
Test for overall effect (random effects): $t_9 = 5.61$ (P < 0.001)

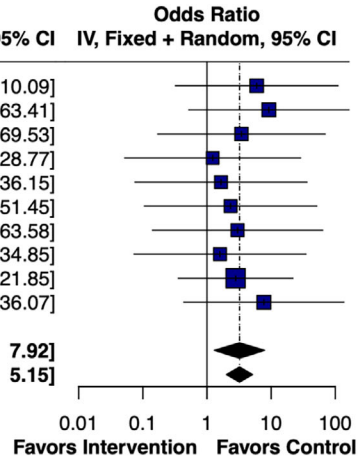

FIGURE 8 | Forest plot of hyperkalemia risk with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

Study	Intervention Events	Intervention Total	Control Events	Control Total	Weight (random)	OR	95% CI
ADVANCED HTN 50mg Stable Dose	8	94	3	48	31.3%	1.40	[0.35; 5.52]
ADVANCED HTN 50mg Adjustment Dose	8	95	3	47	31.3%	1.35	[0.34; 5.34]
TARGET HTN (a) 100mg OD	1	30	0	6	5.4%	0.66	[0.02; 18.13]
TARGET HTN (a) 12.5mg BD	1	22	0	6	5.4%	0.91	[0.03; 25.05]
TARGET HTN (b) 100mg OD	1	31	0	6	5.4%	0.64	[0.02; 17.52]
LAUNCH HTN 50mg Stable Dose	11	538	1	135	14.0%	2.80	[0.36; 21.85]
LAUNCH 50mg Adjustment Dose	7	270	0	135	7.2%	7.71	[0.44; 136.07]
Total (common effect, 95% CI)	**37**	**1080**	**7**	**383**	**.**	**1.55**	**[0.72; 3.34]**
Total (random effect, 95% CI)					**100.0%**	**1.55**	**[0.86; 2.77]**

Heterogeneity: $Tau^2 = 0$; $Chi^2 = 2.21$, df = 6 (P = 0.8995); $I^2 = 0.0\%$
Test for overall effect (random effects): $t_6 = 1.84$ (P = 0.116)

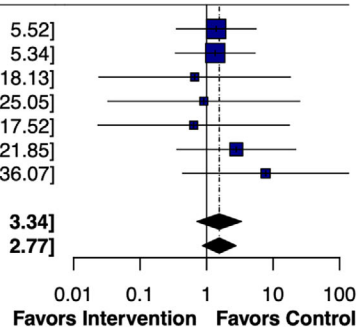

FIGURE 9 | Forest plot of hypotension risk with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

Study	Intervention Events	Intervention Total	Control Events	Control Total	Weight (random)	OR	95% CI
ADVANCED HTN 50mg Stable Dose	8	94	3	48	17.3%	1.40	[0.35; 5.52]
ADVANCED HTN 50mg Adjustment Dose	10	95	3	47	18.2%	1.73	[0.45; 6.59]
LAUNCH HTN 50mg Stable Dose	37	538	5	135	35.9%	1.92	[0.74; 4.98]
LAUNCH 50mg Adjustment Dose	28	270	4	135	28.6%	3.79	[1.30; 11.04]
Total (common effect, 95% CI)	**83**	**997**	**15**	**365**	**.**	**2.16**	**[1.22; 3.83]**
Total (random effect, 95% CI)					**100.0%**	**2.16**	**[1.10; 4.28]**

Heterogeneity: $Tau^2 = 0$; $Chi^2 = 1.62$, df = 3 (P = 0.6557); $I^2 = 0.0\%$
Test for overall effect (random effects): $t_3 = 3.61$ (P = 0.037)

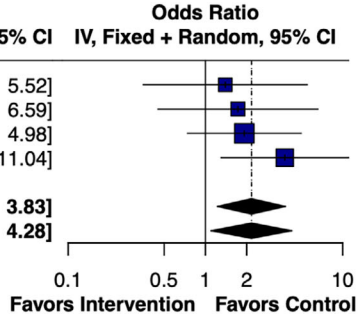

FIGURE 10 | Forest plot of the risk of hyponatremia requiring dose modification with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

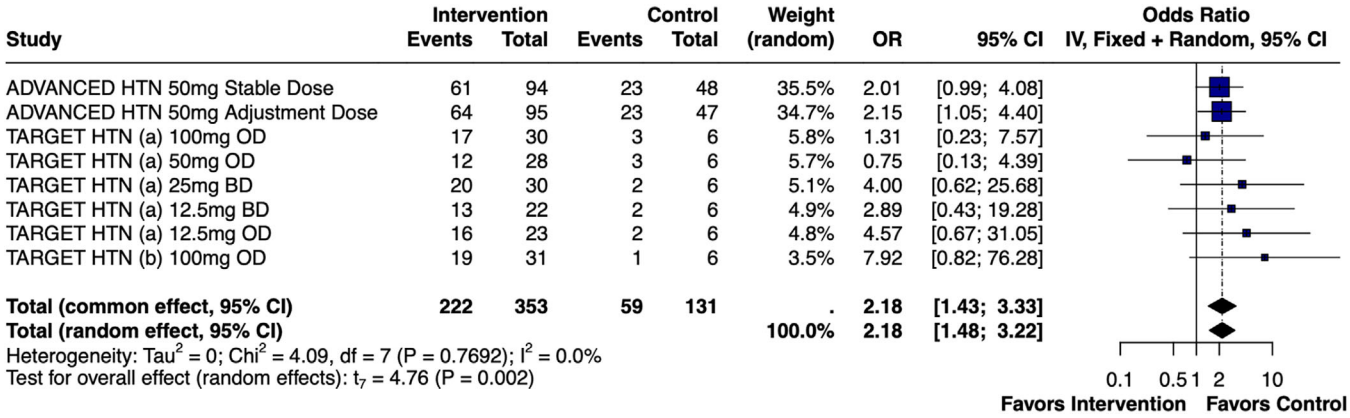

FIGURE 11 | Forest plot of the risk of any adverse event with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

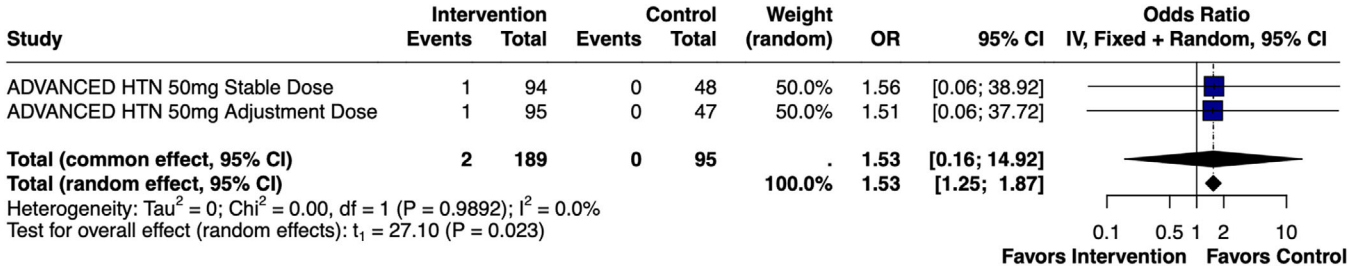

FIGURE 12 | Forest plot of the risk of overdose events with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

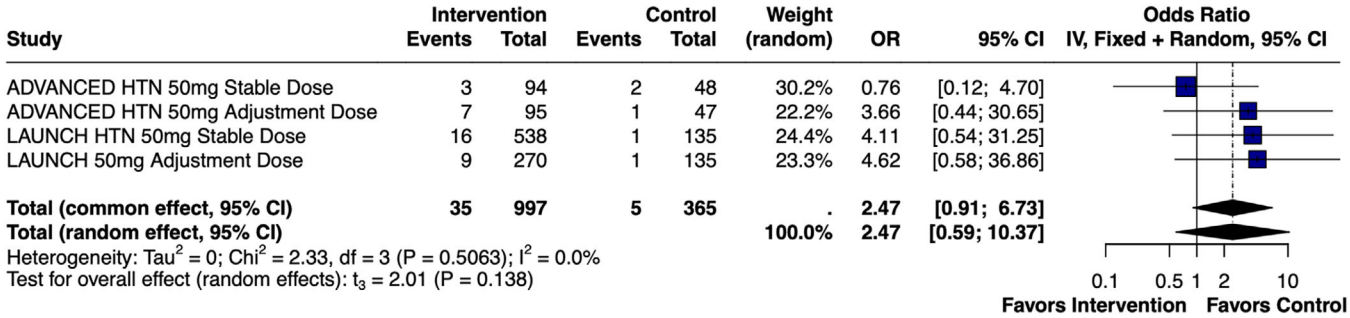

FIGURE 13 | Forest plot of the risk of eGFR reduction requiring dose modification with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

3.2.2.2 | Hypotension. Among the included studies, 7 study arms from 3 RCTs reported hypotension events, comprising 1080 lorundrostat-treated patients (37 events) and 383 control patients (7 events), with no heterogeneity ($I^2 = 0\%$). The analysis showed that the risk of hypotension was comparable between groups and did not reach statistical significance. (OR 1.55; 95% CI: 0.86 to 2.77; $p = 0.116$) (Figure 9).

3.2.2.3 | Hyponatremia requiring Dose Modification. In 4 study arms from 2 RCTs, 83 of 997 lorundrostat-treated patients versus 15 of 365 control patients experienced hyponatremia necessitating dose adjustment, representing a statistically significant increase in risk (OR: 2.16; 95% CI: 1.10 to 4.28; $p = 0.037$) (Figure 10). No heterogeneity was observed ($I^2 = 0\%$).

3.2.2.4 | Any Adverse Events. Across 8 study arms from 2 RCTs, 222 of 353 lorundrostat-treated patients experienced at least one adverse event compared with 59 of 131 controls, corresponding to a statistically significant elevated risk (OR 2.18; 95% CI 1.48 to 3.22; $p = 0.002$) and no heterogeneity ($I^2 = 0\%$) (Figure 11).

3.2.2.5 | Overdose Events. Overdose events were reported in 2 of 189 lorundrostat-treated participants versus none of 95 controls in 2 study arms from 1 RCT, indicating a statistically significant increased risk (OR 1.53; 95% CI 1.25 to 1.87; $p = 0.023$) without heterogeneity ($I^2 = 0\%$) (Figure 12).

3.2.2.6 | eGFR Reduction requiring Dose Modification. In 4 study arms from 2 RCTs, 35 of 997 lorundrostat-treated patients versus 5 of 365 control patients experienced a >25% decline in eGFR from baseline, necessitating dose adjustment. This association did not reach statistical significance (OR 2.47;

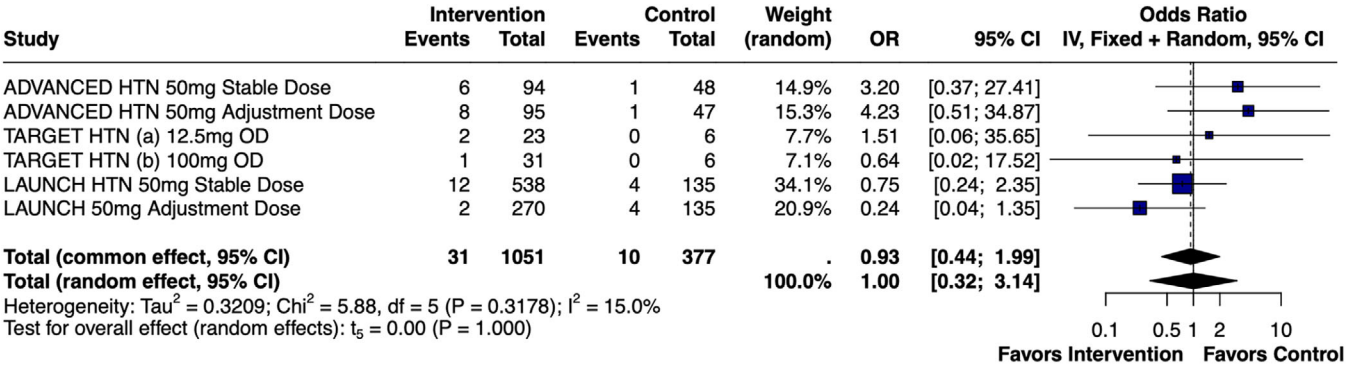

FIGURE 14 | Forest plot of the risk of serious adverse events with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

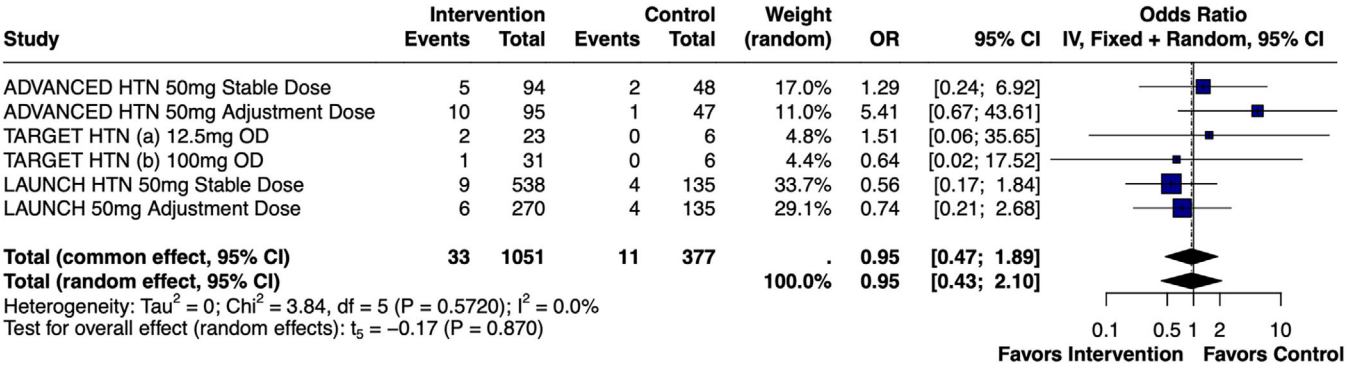

FIGURE 15 | Forest plot of the risk of severe adverse events with lorundrostat versus control. IV, inverse variance; OR, odds ratio.

95% CI: 0.59 to 10.37; $p = 0.138$). No heterogeneity was observed ($I^2 = 0\%$) (Figure 13).

3.2.2.7 | Serious and Severe Adverse Events. Across 6 study arms from 2 RCTs (1,051 lorundrostat-treated vs. 377 control patients), serious adverse events were reported in 31 versus 10 patients (OR 0.93, 95% CI 0.32 to 3.14; $p = 1.00$; $I^2 = 15\%$) (Figure 14), and severe adverse events in 33 versus 11 patients (OR 0.95, 95% CI 0.43–2.10; $p = 0.870$; $I^2 = 0\%$) (Figure 15), indicating no statistically significant difference for either outcome.

3.2.3 | Sensitivity Analyses

We conducted leave-one-out sensitivity analyses for all outcomes to evaluate the influence of individual study arms on the robustness of the pooled effect estimates. In general, these analyses showed consistent results across both primary and adverse outcomes. Notably, the change in office systolic blood pressure from baseline to end of treatment period showed substantial heterogeneity in the main analysis. However, exclusion of a study arm (LAUNCH 50 mg adjustment-dose) reduced heterogeneity to zero ($I^2 = 0\%$) and yielded a statistically significant treatment effect, as shown in Figure 3. Full LOO results for all outcomes are available in Figures S2A–S2I.

3.2.4 | Risk of Bias Assessment

The detailed assessment of 3 included RCTs on ROB-2 showed low risk of bias across all five domains highlighting the high quality of these trials. The detailed traffic light plot and summary plot for risk of bias are illustrated in Figure S1.

3.2.5 | Certainty of Evidence (GRADE assessment)

The certainty of evidence for each outcome was assessed using the GRADE approach. Certainty ranged from high to very low depending on the outcome, with downgrades most often applied for imprecision and low event counts. The detailed GRADE Summary of Evidence table, including footnotes explaining downgrading decisions, is presented in Table S3.

4 | Discussion

Uncontrolled hypertension remains a significant burden for cardiovascular health worldwide, affecting approximately 80% of the hypertensive population [24]. Consequently, the search for new treatment interventions or strategies is paramount. This meta-analysis provides the most comprehensive and quantitative review to date of the efficacy and safety of lorundrostat, a selective aldosterone synthase inhibitor, among adult participants with uncontrolled hypertension.

When the effect on office systolic blood pressure (SBP) was analysed, it was found that the initial analysis of seven study arms pooled from two RCTs had a mean difference of −13.55 mmHg in the lorundrostat versus control group; however, this did not reach statistical significance with high heterogeneity. The LAUNCH 50 mg dose-adjustment cohort accounted for the majority of the heterogeneity. In a sensitivity analysis where this outlier was removed, heterogeneity was eliminated, and the mean difference was significant at −9.08 mmHg. The SBP mean difference in this study approximates the mean BP reduction of 8.7 mmHg in the PATHWAY-2 trial with spironolactone among patients with uncontrolled hypertension despite three antihypertensive drugs, which suggests that lorundrostat may have comparable BP-lowering capability through mineralocorticoid antagonism [25].

Additionally, a mean SBP reduction of 13.29 mmHg was observed in the single-arm continuous analysis of 1067 lorundrostat-treated patients across nine different dosing cohorts, further supporting a dose-responsive antihypertensive effect. These results are consistent with other trials like the TARGET-HTN trial, in which the antihypertensive effects of lorundrostat had also differed considerably with respect to SBP reductions, especially in patients with suppressed plasma renin activity and elevated serum aldosterone levels [26].

The reduction which was observed in ambulatory SBP was equally noteworthy, which is clinically important given that 24-h ambulatory SBP is a better predictor of long-term cardiovascular risk compared with office BP [27]. In the analysis of two study arms of one RCT, lorundrostat was statistically significant for reducing ambulatory SBP by 7.45 mmHg, and heterogeneity was not present. The importance of this clinically relevant ambulatory SBP drug response is notable, as it is known that 24-h BP measurements significantly correlate with left ventricular hypertrophy and cardiovascular events than clinic readings and that the effects of many antihypertensive agents fail to provide sustained 24-h activity [28, 29]. It was also observed that lorundrostat significantly reduced diastolic blood pressure (DBP) by 3.49 mmHg. This reduction corresponds to results from trials with other mineralocorticoid antagonists and has clinical significance since modest reductions in DBP led to a lower risk of stroke, especially in those patients less than 60 years old [30]. Additionally, four study arms reported a significantly lower frequency of severe high BP events in the lorundrostat group, which may reflect a potential role in prevention of hypertensive crises [26].

This meta-analysis also assessed a general range of adverse outcomes. Hyperkalemia was the most obvious safety risk, with the lorundrostat group documenting nearly three times higher risk of hyperkalemia. This was consistent with the findings of the AMBER trial, where potassium-sparing strategies, including spironolactone, were associated with hyperkalemia in chronic kidney disease subjects [31]. Nonetheless, hyperkalemia can be managed with effective monitoring [32, 33]. Patients in the lorundrostat treatment group were also twice more likely to experience clinically relevant hyponatremia warranting dose adjustment. This may be linked to a novel mechanism of aldosterone suppression with subsequent diminished sodium retention consistent with previous mechanistic work on mineralocorticoid antagonists [34]. There was no significant difference in hypotension; however, close monitoring for hypotension in frail or volume-depleted patients is advisable. Adverse events were described to be significantly more common in the lorundrostat group, although these adverse events were mainly mild and well tolerated. In regard to adverse events, while there were only single-digit overdoses, these were statistically higher in lorundrostat-treated patients; this might be the result of prudently unknowingly treating patients of supratherapeutic dosing in titration phases of the medication [35]. The decline in eGFR requiring dose modification did not reach statistical significance, suggesting that the monitoring of renal function is clinically appropriate, with consideration of the safety of lorundrostat relative to placebo on any overt nephrotoxicity profile.

Importantly, there was no statistically significant difference between the lorundrostat and control groups in serious adverse events or severe adverse events, and overall, this indicated the safety of the agent in a high-risk population. When the safety profile was compared to current therapies that patients with uncontrolled hypertension are prescribed (e.g., beta-blockers and central sympatholytics), it appears favorable, particularly since its mechanism of action is targeted to the RAAS with limited off-target hormonal adverse effects [13].

4.1 | Strengths and Limitations

This meta-analysis fills an important void in the current literature by combining all randomized evidence available on lorundrostat use in uncontrolled hypertension. Although studies such as HALO and LOGIC-HTN have already reported positive results of lorundrostat, this analysis synthesizes the findings to make more robust and potentially more broadly generalizable conclusions. The strengths of this study include using data from both ambulatory and office BP, combining the data on adverse events, and attempting sensitivity analyses to assess robustness.

This study has several limitations that warrant consideration. First, the analysis was restricted to three randomized controlled trials. While this inevitably limits statistical power and external validity, the exclusive inclusion of RCTs strengthens internal validity and provides more reliable pooled estimates. Second, all trials had relatively short follow-up periods, which constrains the ability to evaluate long-term mortality and major adverse cardiovascular outcomes. Nevertheless, these durations reflect the current state of available evidence, and the comparable follow-up windows across studies allowed for meaningful short-term comparisons. Third, substantial heterogeneity was observed for some outcomes, particularly office systolic blood pressure, suggesting underlying variability across studies. Although leave-one-out sensitivity analyses were performed, subgroup stratification and meta-regression were not feasible due to the small number of trials. Fourth, the limited number of included studies precluded a formal assessment of publication bias, consistent with Cochrane recommendations. Future research should prioritize adequately powered, long-term randomized trials directly comparing lorundrostat with spironolactone, with particular attention to mortality, major adverse cardiovascular outcomes, and cost-effectiveness to inform clinical practice and guideline implementation.

5 | Conclusion

Lorundrostat emerges as a promising therapy for uncontrolled hypertension. It significantly lowers both office and ambulatory blood pressures in patients with uncontrolled hypertension, and it has a tolerable safety profile with an increased risk for hyperkalemia and hyponatremia. Nevertheless, long-term efficacy and safety validation for resistant hypertension, and in a larger multispecialty cohort with more diversity, needs further investigation to properly integrate its long-term use in clinical practice.

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Ethics Statement

This study is a meta-analysis and did not involve human participants or animal subjects.

Conflicts of Interest

The authors declare no conflicts of interest related to this study. All authors have reviewed and approved the final manuscript.

Data Availability Statement

All data generated or analyzed during this study are included in the manuscript and its supplementary file.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting Figure 1: jch70155-sup-0001-SuppMat.docx